ROYAL COLLEGE OF SURGEONS IN IRELAND

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Module: AI and Machine Learning in Healthcare

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1 Section 1: Model Building

1.1 Exploratory Data Anlysis

The distribution of sex was similar across treatment groups **(Figure 1)**. Although there were substantially more males than females in the dataset, this imbalance was consistent in both the control and acupuncture groups, suggesting no evidence of confounding by sex.

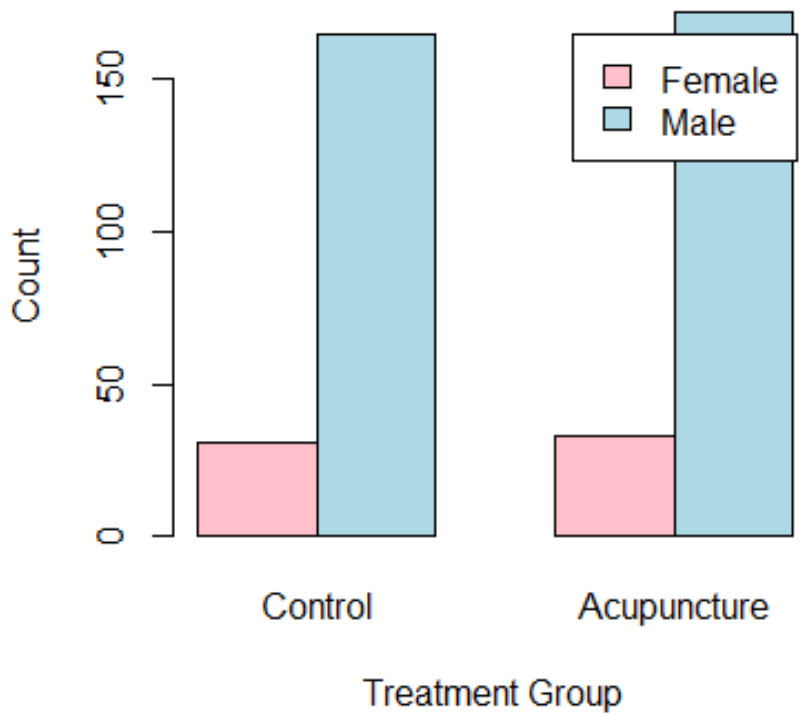

Figure 1: Sex differences by treatment group

The majority of participants were classified as having migraines, which is consistent with the study population **(Figure 2)**. The proportion of migraine and non-migraine patients was similar across treatment groups, indicating no imbalance in migraine status between groups.

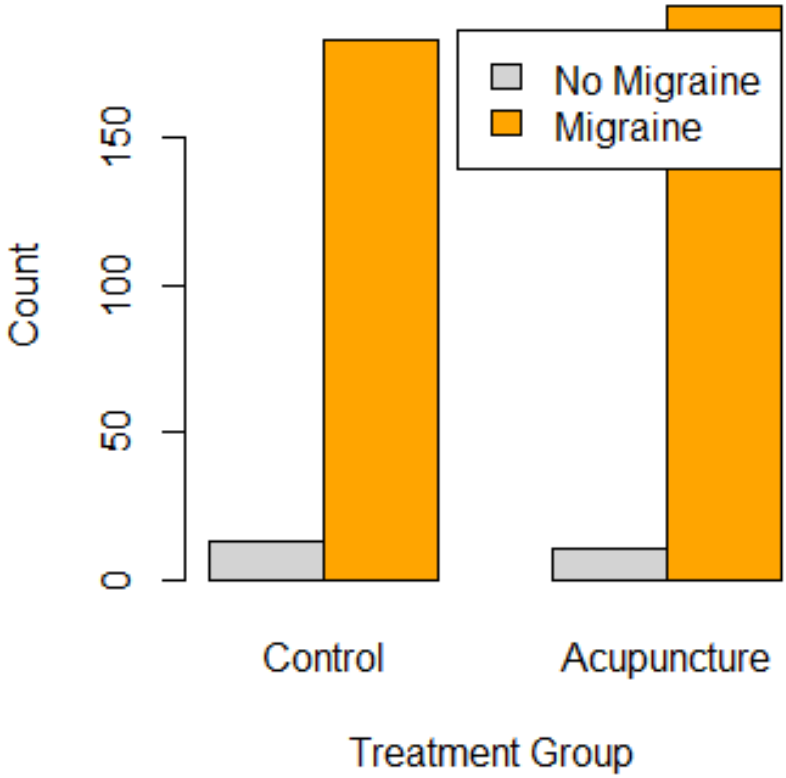

Figure 2: Migraine Differences by Treatment Group

The distribution of age was broadly similar between the control and acupuncture groups, with comparable medians and interquartile ranges **(Figure 3)**. The control group showed slightly greater variability in age, while a small number of younger outliers were observed in the acupuncture group. Additionally, there was no meaningful difference in age distribution between the groups.

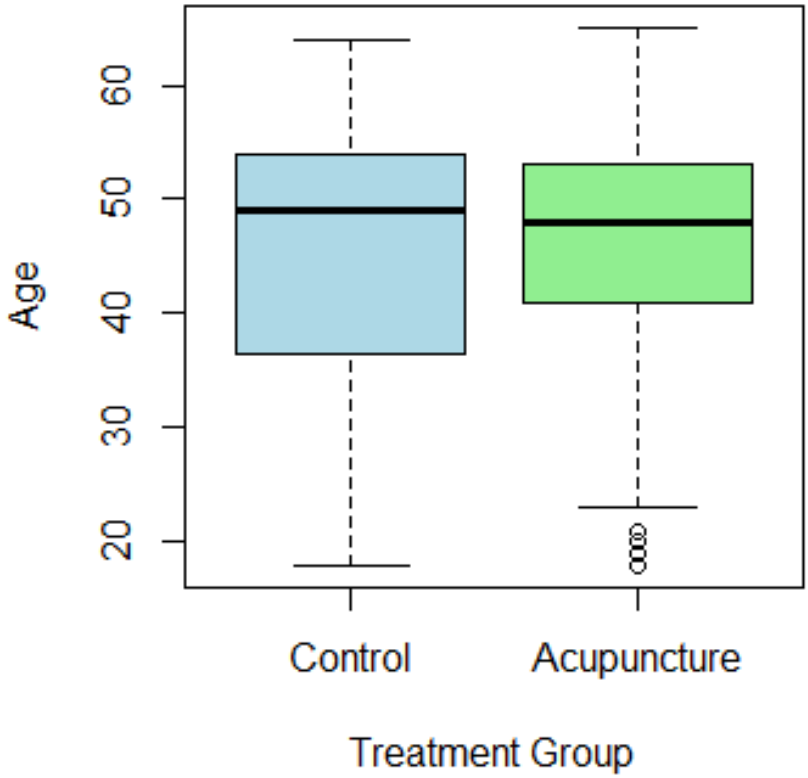

Figure 3: Age Difference by Treatment Group

The duration of headache (chronicity) was similarly distributed across both treatment groups, with comparable medians and spreads **(Figure 4)**. This suggests that participants in both groups had similar histories of headache duration at baseline.

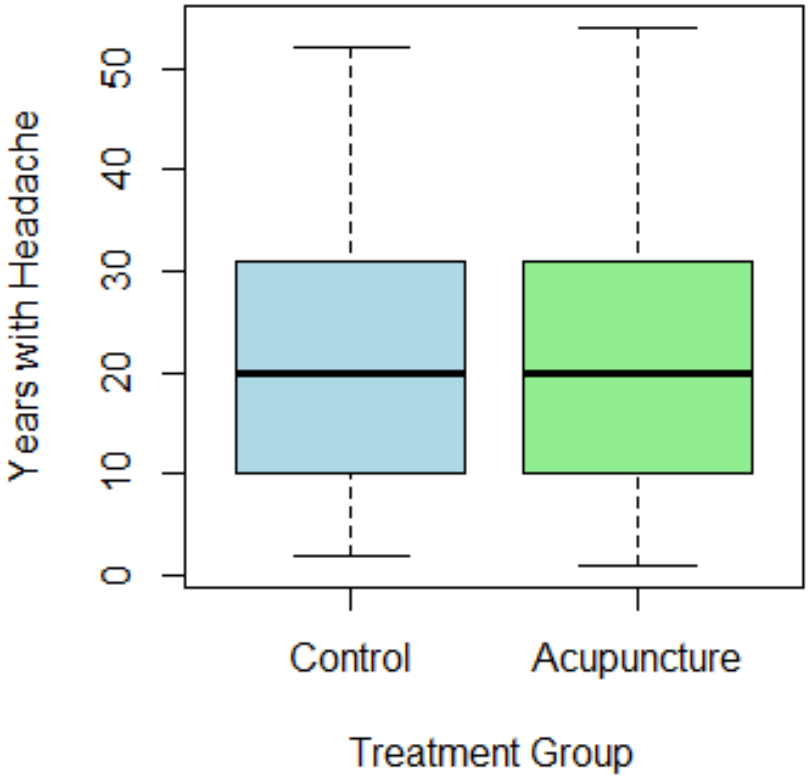

Figure 4: Chronicity Difference by Treatment Group

Summary statistics for key numerical variables are presented in **Table 1**. Participants had a mean age of 45.54 years and long-standing headache duration (mean = 21.46 years). Baseline measures showed moderate variability, with fewer observations available at follow-up due to withdrawals. Pain medication use decreased on average, although variability was high. Overall, baseline characteristics were similar across treatment groups, consistent with the visualisations.

Table 1: Summary statistics for numerical variables

Variable	n	Mean	SD
Age	401	45.54	11.05
Chronicity	401	21.46	13.77
pk1	401	26.51	16.16
pk5	301	19.08	15.61
gen1	400	60.32	22.58
gen5	297	62.21	22.65
painmedspk1	401	63.83	76.60
painmedspk5	301	34.25	49.44
pchange	301	0.21	0.52

1.2 Model Building and Interpretation

A logistic regression model was fitted to assess the effect of acupuncture on treatment response, defined as achieving a $\geq$ 35% improvement from baseline **(Table 2)**. The results indicate that acupuncture was associated with significantly higher odds of response compared to the control group (OR = 2.48, 95% CI: 1.55–3.98, $p < 0.001$). Model fit improved with the inclusion of treatment, as indicated by a reduction in deviance from 412.71 to 397.97. Overall, these findings provide strong evidence that acupuncture is effective in improving outcomes in patients with chronic headaches.

Table 2: Model 1: Effect of acupuncture on treatment response

Term	Coefficient	OR	95% CI	p-value
Intercept	-0.747	0.474	(0.332, 0.675)	< 0.001
Acupuncture	0.909	2.482	(1.550, 3.975)	< 0.001

A logistic regression model was fitted to assess whether the effect of acupuncture on treatment response differed by migraine status **(Table 3)**. The interaction between acupuncture and migraine status was associated with an odds ratio of 7.19 (95% CI: 0.50–103.63, p = 0.147), suggesting a potential difference in treatment effect between migraine and non-migraine patients, however, this effect was not statistically significant. The main effects of acupuncture (OR = 0.38, 95% CI: 0.03–5.17, p = 0.464) and migraine status (OR = 1.45, 95% CI: 0.28–7.48, p = 0.658) were also not statistically significant. Model fit improved with the inclusion of treatment, as indicated by a reduction in deviance from 412.71 to 389.97. Overall, there is insufficient evidence to conclude that the effect of acupuncture differs by migraine status.

Table 3: Model 2: Effect of acupuncture by migraine status

Term	Coefficient	OR	95% CI	p-value
Intercept	-1.099	0.333	(0.067, 1.652)	0.178
Acupuncture	-0.981	0.375	(0.027, 5.169)	0.464
Migraine	0.371	1.449	(0.281, 7.481)	0.658
Acupuncture × Migraine	1.973	7.192	(0.499, 103.635)	0.147

A logistic regression model was fitted to assess whether the effect of acupuncture on treatment response differed by sex **(Table 4)**. The interaction between acupuncture and sex was associated with an odds ratio of 0.81 (95% CI: 0.21–3.14, p = 0.762), indicating no evidence that the effect of acupuncture differs between males and females. The main effect of acupuncture suggested higher odds of response (OR = 3.00, 95% CI: 0.86–10.52), although this was not statistically significant (p = 0.086). There was also no significant effect of sex on treatment response (OR = 1.50, 95% CI: 0.51–4.42, p = 0.462). Model fit improved with the inclusion of treatment, as indicated by a reduction in deviance from 412.71 to 397.18. Overall, there is no evidence that the effect of acupuncture differs by sex.

Table 4: Model 3: Effect of acupuncture by sex

Term	Coefficient	OR	95% CI	p-value
Intercept	-1.099	0.333	(0.121, 0.917)	0.033
Acupuncture	1.099	3.000	(0.856, 10.516)	0.086
Sex	0.405	1.500	(0.509, 4.421)	0.462
Acupuncture × Sex	-0.209	0.811	(0.209, 3.142)	0.762

A linear regression model was fitted to assess the effect of acupuncture on change in general health, defined as the difference between follow-up and baseline scores (gen5-gen1) **(Table 5)**. A positive change in this outcome represents an improvement in general health, while a negative change indicates deterioration.

The results indicate that acupuncture was associated with an average increase of 3.17 units in general health compared to the control group (95% CI: -0.47 to 6.82, p = 0.088). This suggests a potential improvement in general health among patients receiving acupuncture, however, the effect was not statistically significant. The confidence interval includes zero, indicating that the true effect may range from a small decrease to a moderate improvement. Overall, there is insufficient evidence to conclude that acupuncture has a significant effect on general health, although the direction of the effect suggests a possible beneficial trend.

Table 5: Model 4: Effect of acupuncture on change in general health

Term	Coefficient	95% CI	p-value
Intercept	-1.292	(-3.951, 1.367)	0.340
Acupuncture	3.175	(-0.470, 6.820)	0.088

A linear regression model was fitted to assess the effect of acupuncture on change in pain medication use, defined as the difference between follow-up and baseline values (painmedspk5-painmedspk1) (**Table 6**). A negative change in this outcome represents a reduction in medication use, indicating improvement, while a positive change indicates increased medication use.

The results indicate that acupuncture was associated with an average reduction of 9.49 units in pain medication use compared to the control group (95% CI: -20.22 to 1.25, p = 0.083). This suggests a potential decrease in medication use among patients receiving acupuncture, however, the effect was not statistically significant. The confidence interval includes zero, indicating that the true effect may range from a moderate reduction to a small increase in medication use. Overall, there is insufficient evidence to conclude that acupuncture significantly reduces medication use, although the direction of the effect suggests a possible beneficial trend.

Table 6: Model 5: Effect of acupuncture on change in pain medication use

Term	Coefficient	95% CI	p-value
Intercept	-22.521	(-30.374, -14.669)	< 0.001
Acupuncture	-9.485	(-20.222, 1.252)	0.083

1.3 Discussion

Overall, the results suggest that acupuncture is associated with improved outcomes in patients with chronic headaches, particularly in terms of treatment response, where a significant increase in the odds of achieving a ≥ 35% improvement was observed. However, there was no evidence that this effect differed by migraine status or sex. For secondary outcomes, acupuncture showed a trend towards improved general health and reduced medication use, although these effects were not statistically significant.

These findings are broadly consistent with previous research, which has demonstrated that acupuncture can lead to clinically meaningful improvements in headache outcomes, quality of life, and reduced medication use (Vickers et al. 2004). However, in contrast to larger trials, the effects observed in this analysis were weaker and did not reach statistical significance for all outcomes.

A key limitation of this study is the relatively limited sample size available for follow-up analyses, which may have reduced statistical power to detect significant effects, particularly for continuous outcomes. As such, while the direction of effects suggests a potential benefit of acupuncture, larger studies may be required to confirm these findings.

2 Section 2: Decision Trees

A proxy dataframe (proxy_df) was created containing the required variables: ”patient ID”, ”ICD-10”, ”Age”, and ”ATC”. The inclusion and exclusion criteria were implemented using a series of if–else statements, mimicking a decision tree structure.

Firstly, exclusion criteria were applied by checking whether the ICD-10 code began with “F01”, “F02”, or “F03”, corresponding to non-Alzheimer’s dementia diagnoses (*ICD-10 Version:2019* 2026). Patients meeting this condition were excluded. For the remaining individuals, inclusion criteria were then assessed. Patients were included if they had an ICD-10 code beginning with “G30” (indicating Alzheimer’s disease), were aged between 40 and 80 years, and were receiving anti-dementia treatment identified by ATC codes beginning with “N06D” (*ATCDDD - ATC/DDD Index* 2026).

Patients who satisfied all inclusion criteria were appended to a new dataset (Data00), while all others were excluded. This approach reflects a stepwise decision-making process, similar to a decision tree, where each condition acts as a node determining whether a patient is included or excluded from the final cohort.

```python

import pandas as pd

#creating empty proxy dataframe
proxy_df = pd.DataFrame(columns=["name", "age", "group"])

#creating empty list to add rows into
Data00 = []

#creating for loop to go over rows
for index, row in proxy_df.iterrows():
    if row["ICD_10"].startswith(("F01", "F02", "F03")): #exclusion criteria
        pass
    elif row["ICD_10"].startswith("G30") and (40 <= row["Age"] <= 80) and row["ATC"].startswith("N06D"): #inclusion criteria
        Data00.append(row)
    else:
        pass

#creating dataframe with patients having inclusion criteria
Data00 = pd.DataFrame(Data00)
```

3 Section 3: Random Trees and Random Forests

Decision trees are supervised machine learning models that use a series of hierarchical if–else decision rules to partition the data. The structure consists of a root node (initial split), internal nodes (subsequent splits), and terminal leaf nodes representing final class predictions.

The model recursively splits the dataset based on predictor variables, selecting splits that minimise node impurity. For classification trees, impurity is measured using the Gini index. Decision trees are non-parametric and can capture non-linear relationships and interactions between variables, while remaining interpretable.

Random forests are ensemble learning methods based on bagging, where multiple decision trees are trained on bootstrap samples of the dataset. Additional randomness is introduced by selecting a subset of features at each split, reducing correlation between trees and improving generalisation. Predictions from individual trees are aggregated to produce the final classification.

3.1 Random Trees

A decision tree classification model was fitted to predict tumour diagnosis (benign vs malignant) **(Figure 5)**. The resulting tree identified perimeter_worst as the variable at the root node, indicating it is the most important predictor of tumour classification. Subsequent splits were based on concave.points_worst, concave.points_mean, and related features. This suggests that specific geometric characteristics, particularly those related to tumour size and boundary irregularity, are key determinants in distinguishing between benign and malignant cases.

This is supported by the literature, where malignant tumours are characterised by invasive growth, increased cell mobility, and structural irregularity, all of which contribute to complex and heterogeneous tumour geometry (Baba and Câtoi 2007). In contrast to benign tumours, which are typically well-circumscribed and exhibit expansive growth, malignant tumours display infiltrative growth patterns with irregular extensions into surrounding tissue (Grizzi 2015). Features such as concave points capture this boundary irregularity, reflecting the ragged tumour–host interface that is a hallmark of malignancy (Baish and Jain 2000). Furthermore, tumour geometry has been shown to directly relate to tumour aggressiveness and metastatic potential, with highly irregular, non-spherical shapes indicating more invasive behaviour (Almagro et al. 2022).

The model correctly classified 79 benign and 48 malignant cases, with 10 benign tumours incorrectly classified as malignant and 5 malignant tumours incorrectly classified as benign. Overall, the model demonstrates good predictive performance with relatively few misclassifications **(Figure 7)**. The balanced accuracy of the decision tree model was 0.897, indicating strong overall classification performance across both classes (Bobbitt 2021).

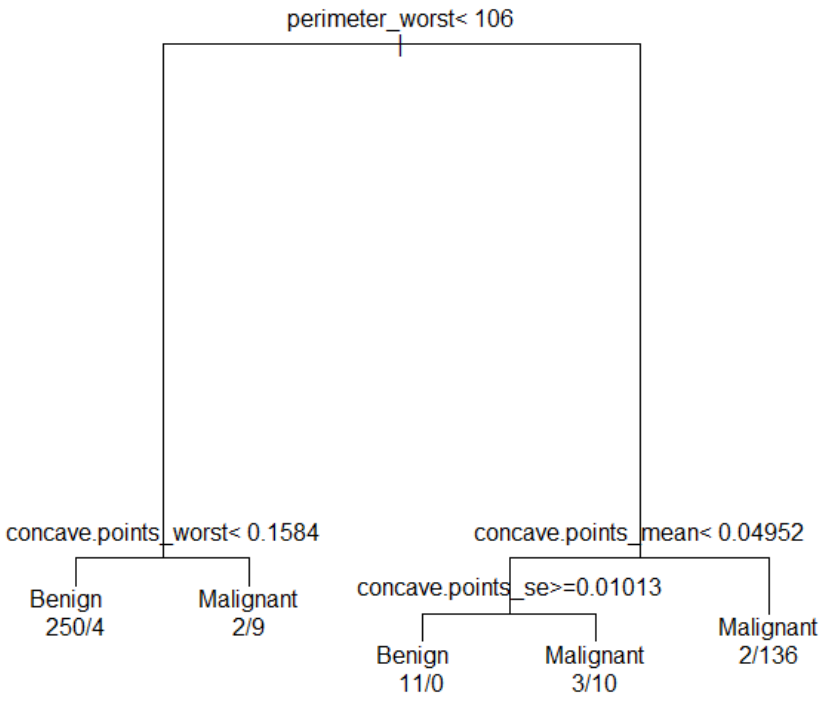

Figure 5: Random Tree

Actual / Predicted	Benign	Malignant
Benign	79	10
Malignant	5	48

Table 7: Confusion matrix showing classification performance of the Random Forest model

3.2 Random Forests

A random forest model was fitted as an extension of the decision tree approach. The random forest model correctly classified 84 benign and 48 malignant cases, with 5 benign cases incorrectly classified as malignant and 5 malignant cases incorrectly classified as benign **(Table 8)**. Compared to the decision tree, the random forest reduced the number of misclassified benign cases, resulting in improved overall classification performance.

The balanced accuracy of the random forest model was 0.925, compared to 0.897 for the decision tree. This indicates that the random forest provides improved predictive performance (Bobbitt 2021). The improvement is likely due to the ensemble approach, which reduces overfitting and captures more complex patterns in the data compared to a single decision tree.

Overall, while the decision tree provides an interpretable model with clear decision rules, the random forest achieves higher accuracy and more robust predictions, making it better suited for classification performance.

Actual / Predicted	Benign	Malignant
Benign	84	5
Malignant	5	48

Table 8: Confusion matrix for Random Forest model

3.3 Feature Comparison

The decision tree model identified perimeter_worst as the most important variable, followed by area_worst, radius_worst, perimeter_mean, and area_mean **(Table 9)**. The importance of perimeter_worst is consistent with its position as the root node, indicating that it provides the most informative initial split in the dataset. This suggests that overall tumour size is prioritised for the primary split, with further refinement based on related size measures

Variable	Importance
perimeter_worst	146.02
area_worst	142.14
radius_worst	142.14
perimeter_mean	123.49
area_mean	119.88

Table 9: Top 5 important variables from the Decision Tree model

The top variables identified by the random forest model included concave.points_worst, area_worst, radius_worst, perimeter_worst, and area_se, all of which showed high values for both mean decrease in accuracy and mean decrease in Gini **(Table 10)**. These results indicate that features capturing ”worst” tumour characteristics and boundary irregularity contribute most to classification performance, suggesting that the most pronounced geometric deviations are the most informative for distinguishing between benign and malignant cases.

Variable	Benign	Malignant	Mean Decrease Accuracy	Mean Decrease Gini
concave.points_worst	6.37	4.72	7.30	23.97
area_worst	6.03	5.69	7.78	25.07
radius_worst	5.78	4.91	8.00	15.06
perimeter_worst	5.23	5.10	6.63	20.47
area_se	4.80	2.84	5.69	12.99

Table 10: Top 5 most important variables from the Random Forest model

3.4 Discussion

These results align with biological observations that malignant cells exhibit accelerated growth, genomic alterations, and morphological abnormalities, including enlarged and irregular structures that contribute to increased tumour size and altered geometry (Baba and Câtoi 2007). However, size alone does not fully capture malignancy, as the spatial organisation and structural complexity of the tumour are also critical (Baish and Jain 2000).

The decision tree results suggest that tumour size provides a strong initial distinction between benign and malignant cases. In contrast, the random forest places greater emphasis on boundary irregularity, with concave.points_worst emerging as a key feature. This indicates that while size is an important discriminator, more subtle geometric characteristics related to tumour shape and boundary complexity provide additional discriminatory power.

This is consistent with the defining characteristic of malignant tumours as invasive and infiltrative, where irregular growth patterns produce complex, fractal-like geometries and irregular tumour–host interfaces (Grizzi 2015). Such geometric complexity reflects underlying biological processes including increased cell motility and tumour–host interaction, which facilitate invasion and metastasis (Almagro et al. 2022).

The difference in variable ranking between the models reflects their underlying mechanisms. A decision tree looks for the single best split at each step, so a small number of variables end up dominating the model. In contrast, the random forest builds many trees using different samples and subsets of features, so importance is based on how much each variable contributes across all of these trees. This results in a broader and more reliable assessment of which features are actually important overall.

References

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